

A Packing of 12 Four-Spaces in $\mathbb{R}^{11}$

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Abstract

We give 12 4-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 2.776562. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.757240, obtained by deleting 1 of the 13 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 12 subspaces of dimension 4 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.776562 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 4$. The cell $N = 12$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.757240, obtained by deleting 1 of the 13 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(11 - 4)/11 \cdot 12/11 = 2.776860$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N12.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 62 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
308506903015121855004816467361962349768062257532523131711937484401822359
531693684117641215304837032783444857101985305697897171192046541459596858
202933403822972085580739308841280377980417940303
```

denominator:

```
111111111111235510782967462322549591329613300360329521319707528927265684
470589398006064993896382082601523956099645371820486623301074558318053668
846990978087407085879232604237536578569793042117
```

B Coordinates

12 blocks of 11 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m11n4N12.txt`.

```
-553374940097 -425584854090 45037049622 -3850089302257409213711 -123616819322 224212062973 1787081618463525025267 -123196721633 -212449487746 373237864778
72408372576 -384099276903 -57994681084 -2228896522380175189073 497946692965 121637434696 445376777540-111004879347 -205954183407 -71798918499 -261025567202
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